

SurakshaNet: on-device Hinglish abuse detection

B.Tech mid-semester briefing

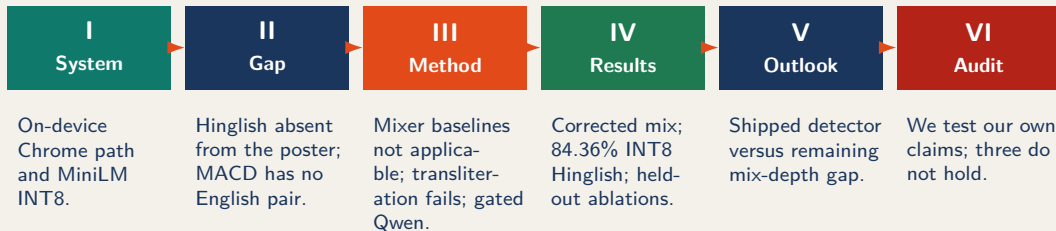
Aamod Jain and Pratham Garg · 20 September 2026

CHROME MV3 INFERENCE



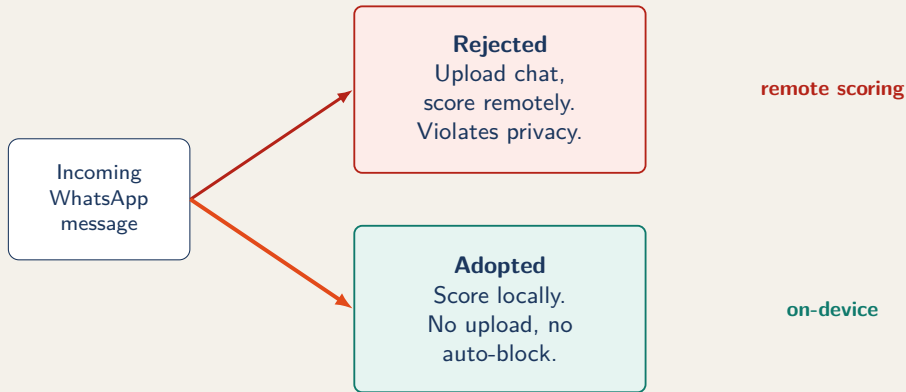
Conversion is training-time. Inference never calls Qwen.

Overview



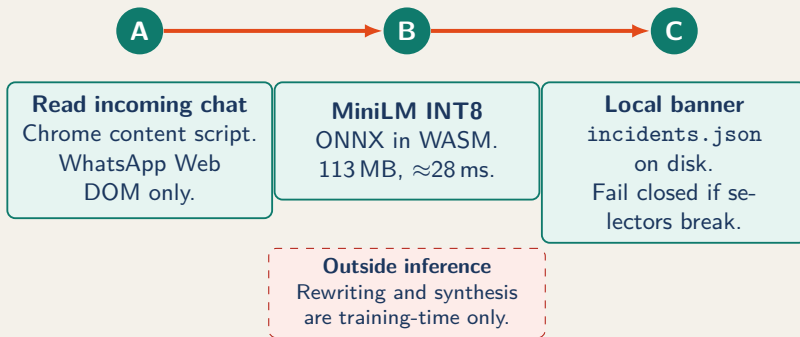
Inference remains the shipped MiniLM path. The Hinglish contribution is a training-set construction that does not alter A–B–C.

On-device scoring; a cloud round-trip is not a valid design.



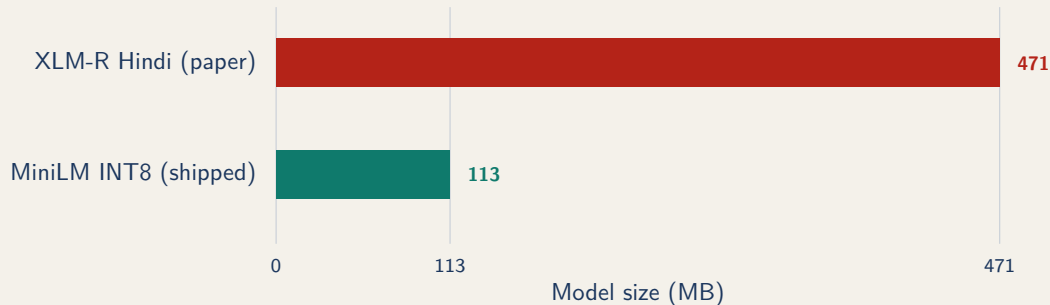
Term — on-device. *The model, the score, and incidents.json remain on the user's laptop.*

Inference path A–B–C; conversion is training-time only.



Term — Chrome MV3. *Current extension platform. Caps bundle size and forbids a silent cloud round-trip at inference.*

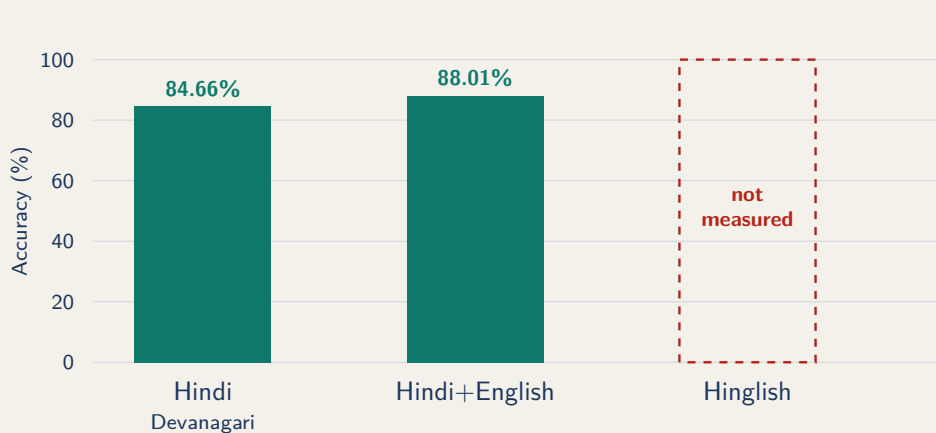
XLM-R exceeds the Chrome size budget.



MACD XLM-R Hindi ceiling is 86.34%. MiniLM INT8: poster Hindi 84.66%, drop under 0.11 pp, 28 ms.

Term — INT8. 8-bit quantized ONNX that Chrome loads. FP32 is the training checkpoint.

Poster evaluation omitted Hinglish.



The dashed bar is a missing measurement, not a score. Subsequent slides fill that cell without changing A-B-C.

Labeled MACD is Devanagari; traffic is Roman Hinglish.



Term — Hinglish. *Hindi as the matrix language, written in Roman letters, with English words mixed in. Not ITRANS, and not Devanagari in another font.*

Training uses published MACD and Davidson labels.



Term — MACD. *Multilingual Abusive Comment Detection. ShareChat-scale Hindi comments with published labels.*

Research question

Can MACD Hindi be rewritten as conversational Hinglish with published labels preserved, then measured on a test frozen before training?

Constraint 1

Inference path A-B-C is unchanged.

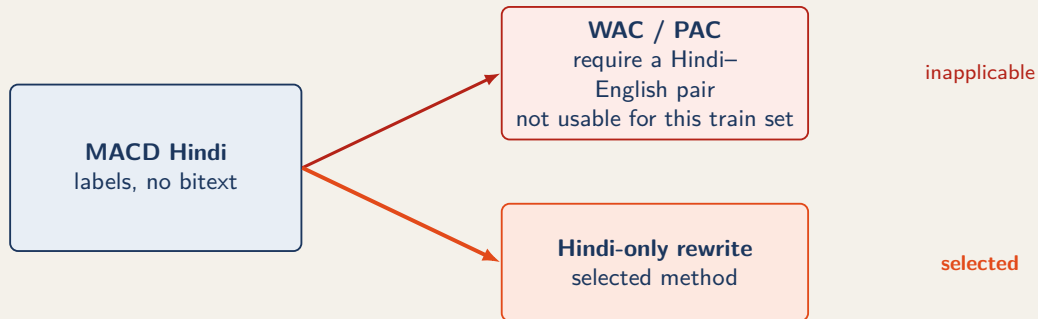
Constraint 2

Class 0 remains abusive after rewrite.

Constraint 3

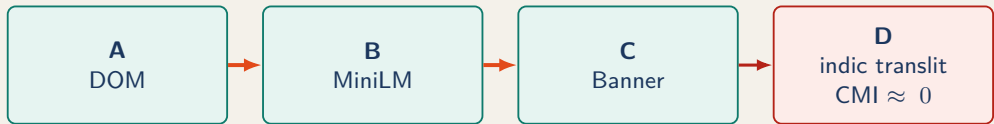
Hinglish test $n = 6368$ is frozen first.

WAC and PAC require bitext; MACD provides Hindi only.



Term — HinGE / WAC / PAC. *HinGE: human Hinglish benchmark, $n \approx 1964$ (IITGN). WAC: word-aligned mixing. PAC: phrase-aligned mixing. Both require bitext. HinGE is used later as converter evaluation, not as a train-set builder.*

Script transliteration is not code-mixing.



rejected as a training method

Hypothesis. Cheap, no GPU. If the language mixed, CMI would rise and WER against ITRANS would stay low.

Result. The sentence is still Hindi in another script. Script conversion $\neq$ code-mixing.

Term — CMI. *Code-mixing index (Das & Gambäck). 0 = one language. Higher = deeper mix.*

Term — WER. *Word error rate $(S + D + I)/N$. Low WER vs ITRANS means “just transliteration.”*

Only conversational Hinglish matches the target register.

1. MACD Hindi

कल मीटिंग है चार बजे

Labeled corpus.
Devanagari. Gold class.

2. indic_transliteration

kala mITiMga
hai cAra baje

Rejected. Same
Hindi. CMI $\approx$ 0.

3. Conversational mix

kal meeting hai
4 baje

Target. English words,
Hindi grammar.

Qwen3-32B rewrite is offline; it is not in Chrome inference.

INFERENCE — Qwen is not called

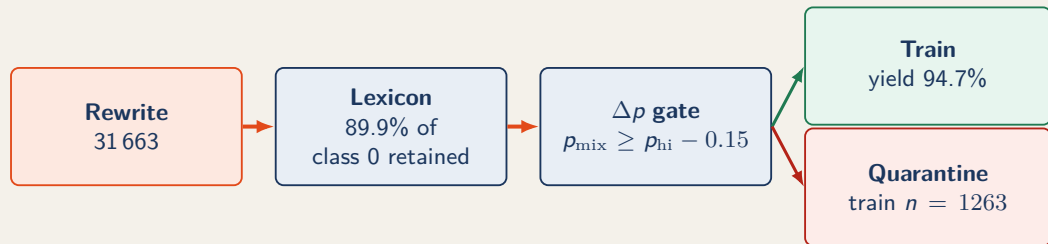


TRAINING — Bedrock Mantle, us-east-1, once



31 663 lines converted. Qwen is a converter, not a detector, and it is not inside Chrome.

Quarantine preserves class-0 labels after sanitisation.



Term — quarantine. *Held-out rejected rewrites with a reason: score drop, missing lexicon stem, collapsed length, refusal, leftover Devanagari. Not deleted. Not in train.*

Hinglish evaluation is frozen at $n = 6368$ before training.

Protocol

Train cleaning is allowed.
Quarantine is logged, not deleted.
Synthetic rows stay out of val/test.
CMI is reported even when low.

Forbidden

Rewriting the frozen 6 368 after error analysis.
Shipping a nearby ablation.
Placing Qwen inside Chrome.

Moving the test after error analysis would make 84.36% incomparable to the poster. Layer-1 cleaning and 977 synthetic rows were tried on train only.

Omitting Hindi MACD collapses Devanagari accuracy to 70.04%.

Failed mix

Train = Davidson 80% + Hinglish only.
Hindi test 70.04% ($n = 6728$).
Poster Hindi accuracy collapsed.

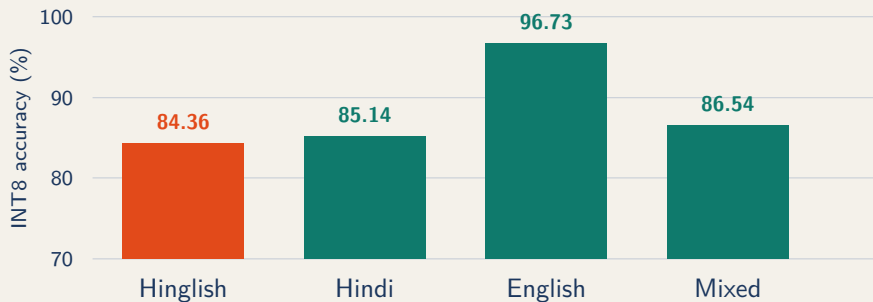


Corrected protocol

Train = Hindi MACD +
Davidson 80% + gated Hinglish.
Hindi INT8 85.14%. Shipped mix.

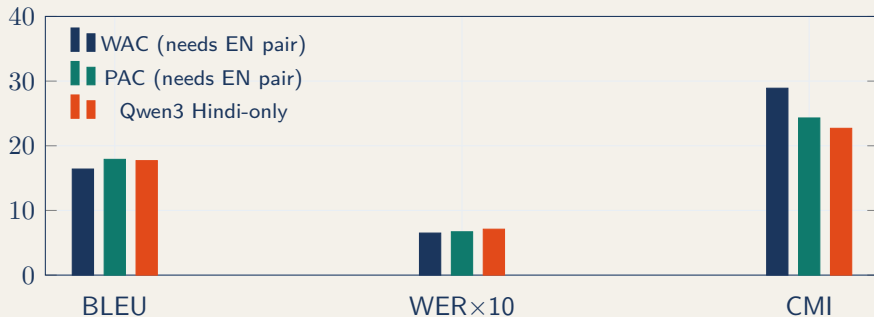
Corrected sizes: train 58 929 / val 15 581 / test 15 575. A Hinglish gain that costs Hindi is not adopted.

Shipped INT8 Hinglish accuracy is 84.36% with A–B–C unchanged.



Frozen Hinglish $n = 6368$. Mixed 86.54% is lower than poster 88.01% because Hinglish is now inside the mix. Same MiniLM, 113 MB, 28 ms.

HinGE: Hindi-only Qwen versus WAC and PAC

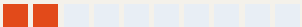


Human HinGE CMI = 35.6. This table evaluates the *converter*. It does not replace detector accuracy. **Term** — **BLEU**. Precision against a human Hinglish reference. Not abuse-detection accuracy.

MACD rewrites mix more shallowly than HinGE human references.

$$\text{CMI} = 100 \left(1 - \frac{\max_i w_i}{n} \right)$$

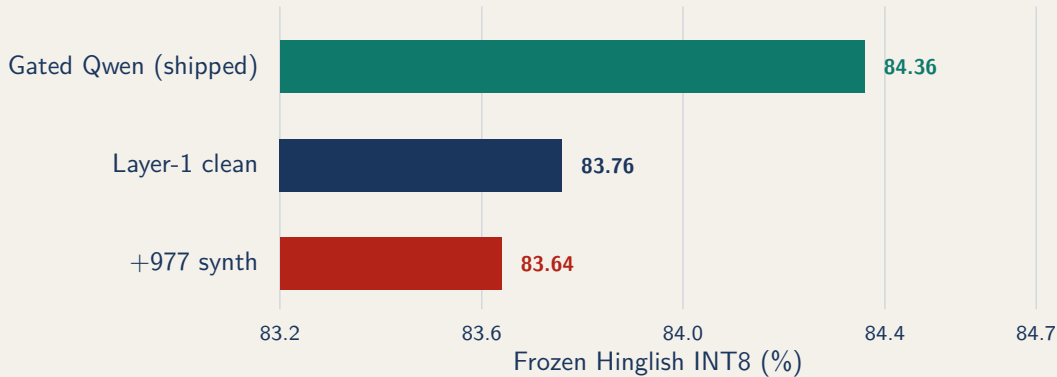
n : language-tagged tokens (no emojis). w_i : count in language i . CMI = 0 if only one language remains.

Qwen MACD  13.7

Humans  35.6

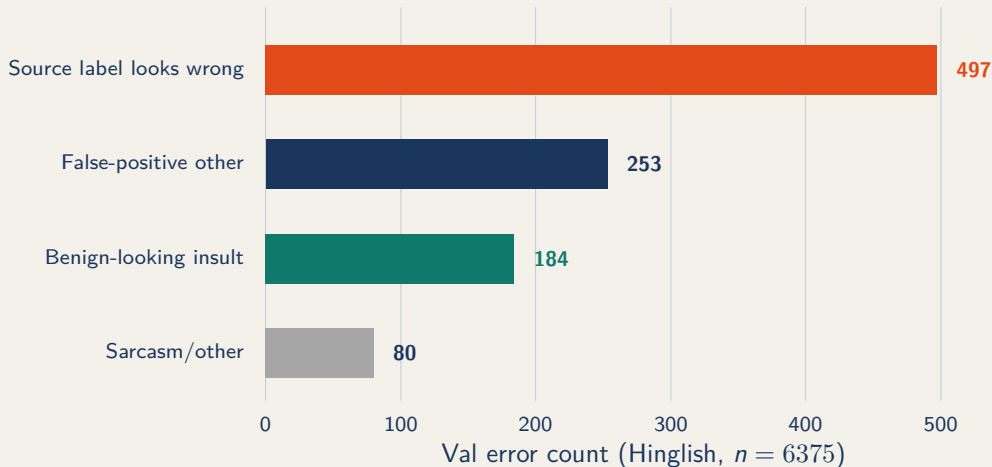
25% of Qwen MACD lines have CMI = 0 (romanized Hindi or English, not a mix).

Layer-1 and synthetic rows do not improve the freeze.



Deltas -0.60 and -0.72 on the same freeze. Synthetic rows never entered test. The nearby recipes are not shipped.

Largest validation errors are noisy MACD gold, not obfuscation.



507 false positives, 528 missed abuse. Cluster labels are a research reading, not a second gold. Cleaning this on the *test* set is exactly what the freeze forbids.

Where we stood before auditing our own work.

Believed closed

On-device A–B–C, 113 MB, 28 ms.

Hinglish 84.36% on a frozen test.

Hindi recovered after 70.04%.

Gated Qwen; transliteration and synth rejected.

Open

Raise CMI toward HinGE humans (35.6)

without sanitising class 0,

without unfreezing $n = 6368$,

and without placing Qwen in Chrome.

Part VI re-examines the left-hand column against the artifacts on disk. Only the first line survives unchanged; three of the others do not. Raising CMI stays the right move, but it has to be judged on a different metric than the one implied here.

We audited our own results. Three of them did not hold.

Every number in this section is recomputed from artifacts on disk, not carried over from the earlier slides. Scripts: `aws_train/audit_*.py`, `aws_train/gate_v*.py`.

What the audit asked, and what it found.

1. One number, one blind spot

Is 84.36% informative?

No. Abuse recall falls
87.4 → 67.1% with mixing
depth while accuracy stays flat.

2. The gate was never measured

Where did “89.9% re-
tained” come from?

It is the gate’s own pass rate.
Against real ground truth: AUC
0.63 on 64.6% of abusive rows.

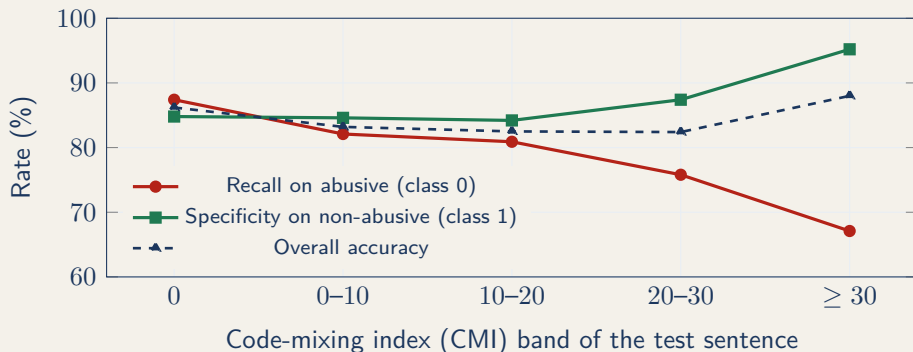
3. Verdicts drawn without a test

Is Layer-1 really worse?

Not established. McNemar
 $p = 0.179$, so the -0.60 pp “do
not ship” verdict is withdrawn.

The inference path A–B–C is unchanged. What changes is what we are entitled to claim about it.

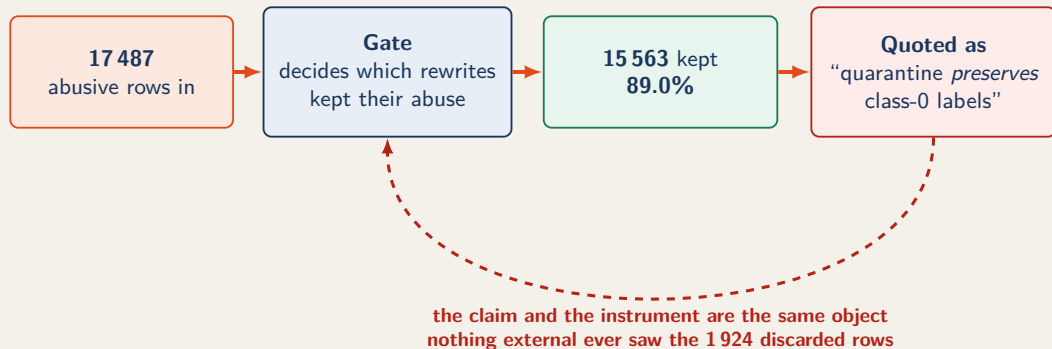
Issue 1: abuse recall collapses with mixing depth.



Shipped INT8, frozen Hinglish test $n = 6368$. Cochran–Armitage trend on recall: $\chi^2 = 52.0$, $p = 5.5 \times 10^{-13}$. Shallow (CMI < 10) recall 85.1% versus deep (CMI ≥ 20) 72.4%, Fisher $p = 2.3 \times 10^{-10}$.

Why accuracy hides it. Deep mixes are mostly non-abusive — the abusive share falls 54.7% $\rightarrow$ 25.6% across these bands — so a model drifting toward “non-abusive” scores *well* there while missing a third of real abuse.

Issue 2: the retention figure is the gate's own pass rate.



It is also mis-attributed. The earlier slide puts the figure under a *Lexicon* box, but of the 1 924 discards the word list made **74** and the Δp score made **1 806** — 94% of the decisions came from the component that diagram never names.

The threshold was never derived. $\delta = 0.15$ is an `argparse` default, reused unchanged for a second, different gate. A repository-wide search finds no human annotation, no threshold sweep, no sensitivity analysis. Where detection can be scored at all, Youden's J puts the optimum at 0.019 — roughly eight times stricter.

What we did: generate ground truth. Two designs failed first.

Attempt 1 — invalid

Mine the word list automatically, delete one slur occurrence.

24.3% of “abuse removed” rows still contained a slur; neutral words such as *kapde* were deleted.

Attempt 2 — tautological

Require exactly one hard slur aligned to a Devanagari stem, with post-conditions.

Labels sound, but deletion leaves a broken fragment and the word check *is* “is that token gone”: it scores 100%, an LLM judge only 10.3%.

Attempt 3 — used

Negatives are *fluent* softened rewrites from the same Qwen3-32B; positives are fluent faithful rewrites.

Both arms natural text, so no gate wins by spotting bad grammar.

4 800 graded items over 1 200 sources for the covered stratum, 466 over 233 sources for the blind stratum, $\approx$ 2 100 Bedrock generation and judging calls. Remaining gap, stated plainly: labels come from the generation instruction and are verified only lexically. Human certification of a subsample is Stage A of the next iteration.

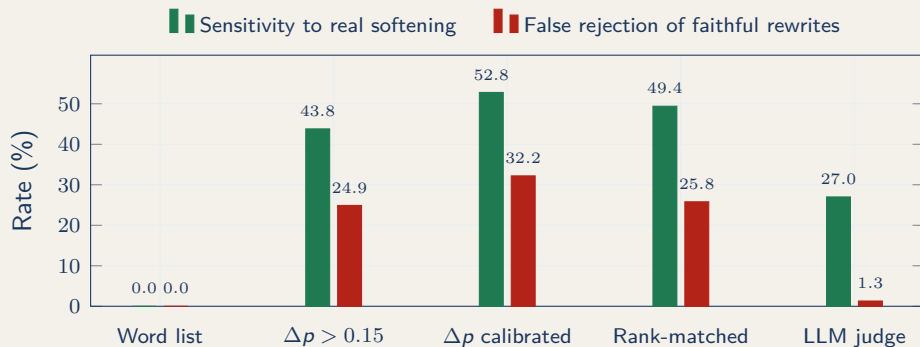
Metric: the gate splits into two strata that need different tools.

	Word list applies 35.4% of abusive rows	Word list vacuous 64.6% of abusive rows
Lexicon check, sensitivity	99.8%	0.0%
Lexicon check, false rejection	0.0%	0.0%
$\Delta p > 0.15$, sensitivity	48.0%	43.8%
$\Delta p > 0.15$, false rejection	1.6%	24.9%
AUC of Δp vs real softening	0.834	0.629

Left column. Where the 20 stems fire, word-level checking is essentially perfect and the Δp score contributes nothing but false rejections. A word list *is* sufficient here.

Right column. On the majority of abusive rows `lexicon_preserved` returns true because nothing was required, so Δp decides alone — at AUC 0.629, where chance is 0.500.

Metric: no automatic substitute reached a usable operating point.



Lexicon-blind stratum, 466 items from 233 sources. **Calibrating** the threshold buys 9 pp of sensitivity for 7 pp of false rejection. **Rank-matching** the two probabilities so both are read in the same script — our first proposed fix — moves AUC $0.629 \rightarrow 0.636$, i.e. nothing, so the defect is not calibration. **An LLM judge** shown the Devanagari source is far more precise but catches only 27.0%, and is not significantly better overall than the shipped rule (McNemar $p = 0.21$).

Issue 3: ship/no-ship decisions needed a paired test.

Comparison	Shipped	Variant	Diff	p (McNemar)	Verdict
shipped vs Layer-1	84.28%	83.81%	+0.47 pp	0.179	not significant
shipped vs Layer-1 + synth	84.28%	83.46%	+0.82 pp	0.027	significant

Selection ran on the frozen test

`_int8_hinglish()` reads `int8.test_hinglish.accuracy`; the best mix is picked by it and promoted if it beats the extension. 84.36% is a selected-on-test estimate.

Neither artifact is reproducible

The shipped recipe matches no `MIX_CONFIG` entry (`mix: None`), and promotion overwrote the checkpoint that gated the corpus.

Re-running the bundles reproduces the logged accuracies to within 0.18 pp, so padding and batching alone move the metric as much as the verdicts drawn from it.

Two claims of our own are withdrawn.

Withdrawn: “the gate penalises code-mixing”

On the invalid first benchmark, injecting benign English words appeared to raise rejection odds significantly. On the corrected benchmark the same injection lifts

CMI 9.98 $\rightarrow$ 26.99 and rejection *falls*,
2.00 $\rightarrow$ 1.25%: paired odds 0.61, McNemar
 $p = 0.19$. The direction is opposite to the claim.

Withdrawn: the “~14% label noise” estimate

A Rogan–Gladen inversion of the 11% quarantine rate. Invalid: it used the broken benchmark and ignored `max_retries`, so the observed rate is not a single-test positive rate.

What replaces the second one. With sensitivity and false rejection measured, contamination of the accepted corpus is $s(1 - \text{TPR}) / [s(1 - \text{TPR}) + (1 - s)(1 - \text{FPR})]$, reported as a sweep over the unmeasured softening rate s rather than as a point estimate. At $s = 20\%$ the gate takes contamination from 20% to 15.8% — it removes about a fifth of what it exists to remove, while discarding 1924 abusive rows outright.

The surviving observational claim is weak and we state it as such: quarantined rewrites mix slightly more deeply than kept ones, CMI 12.24 versus 11.28, Mann–Whitney $p = 0.014$, and a control shows most of it tracks source difficulty.

Conclusion: the claim we can defend, and the number to beat.

Established

On-device A-B-C unchanged: 113 MB, ≈ 28 ms.

Code-mix *depth*, not Hinglish as such, is what breaks detection: recall $87.4 \rightarrow 67.1\%$, $p = 5.5 \times 10^{-13}$.

The gate that built the corpus runs at AUC 0.629 on 64.6% of abusive rows, and no automatic substitute we tested repairs it.

Next iteration

A Human-label 400 rewrites, report κ — needed because every automatic gate failed.

B Rebuild per stratum: drop Δp where the word list fires, combine judge precision with score recall where it does not.

C Generate a depth-balanced abusive corpus.

D Re-freeze a depth-stratified test set; McNemar for every ship decision.

Target for the final report. Recall at $\text{CMI} \geq 20$ from **72.4%** to $\geq 82\%$, specificity not below 84%. Not accuracy: raising CMI does nothing for accuracy (Fisher $p = 0.73$).